MTRN3100

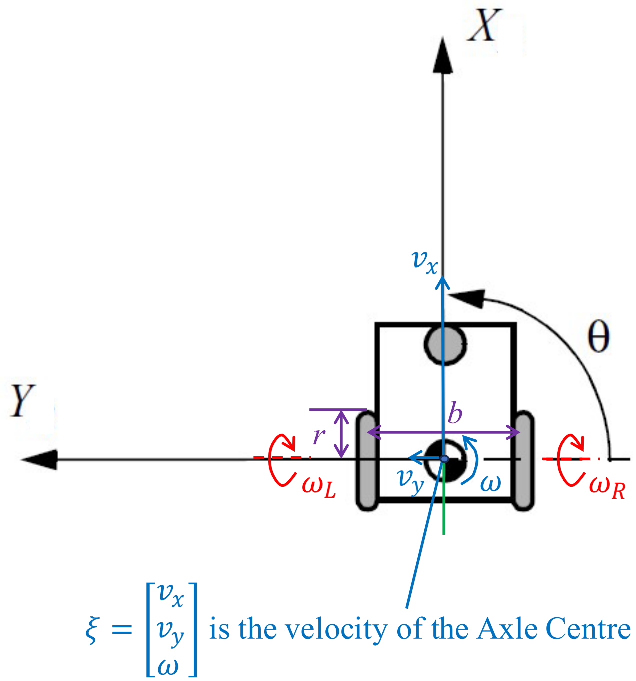
Tutorial 4 – Odometry

Kinematic Model

PLEASE NOTE:
These tutorial slides are a shortened version of the lecture. Watch the lecture for a deeper understanding.

Differential Drive

Two wheeled robot. Drives straight if both wheels turn at the same rate. Turns if the wheels turn at different rates.



Has two parameters:

- r – Wheel radius
- b – Axle length

Kinematic Model

A kinematic model provides a mathematical representation for a system. For the case of a differential drive, the kinematic model can relate wheel position to a global pose.

Increment

Current pose

Next pose

$$p(t+\Delta t) \approx p(t) + \begin{bmatrix} \Delta s \cdot \cos(\theta) \\ \Delta s \cdot \sin(\theta) \\ \Delta\theta \end{bmatrix}$$

$\Delta s \equiv \dfrac{r \cdot \Delta\theta_L}{2} + \dfrac{r \cdot \Delta\theta_R}{2}$ — Incremental linear motion

$\Delta\theta \equiv -\dfrac{r \cdot \Delta\theta_L}{b} + \dfrac{r \cdot \Delta\theta_R}{b}$ — Incremental rotation

$\Delta\theta_L = \omega_L \cdot \Delta t$ — Incremental rotation of left wheel

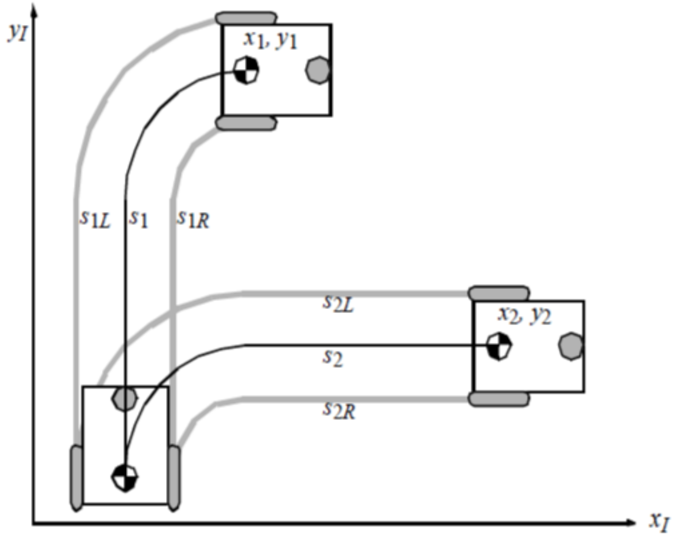
$\Delta\theta_R = \omega_R \cdot \Delta t$ — Incremental rotation of right wheel

Inverse Kinematics

Inverse kinematics is the process of calculating a kinematic model input provided with the desired output.

Can be achieved by re-arrange the linear kinematic model to solve for wheel speed or encoder counts.

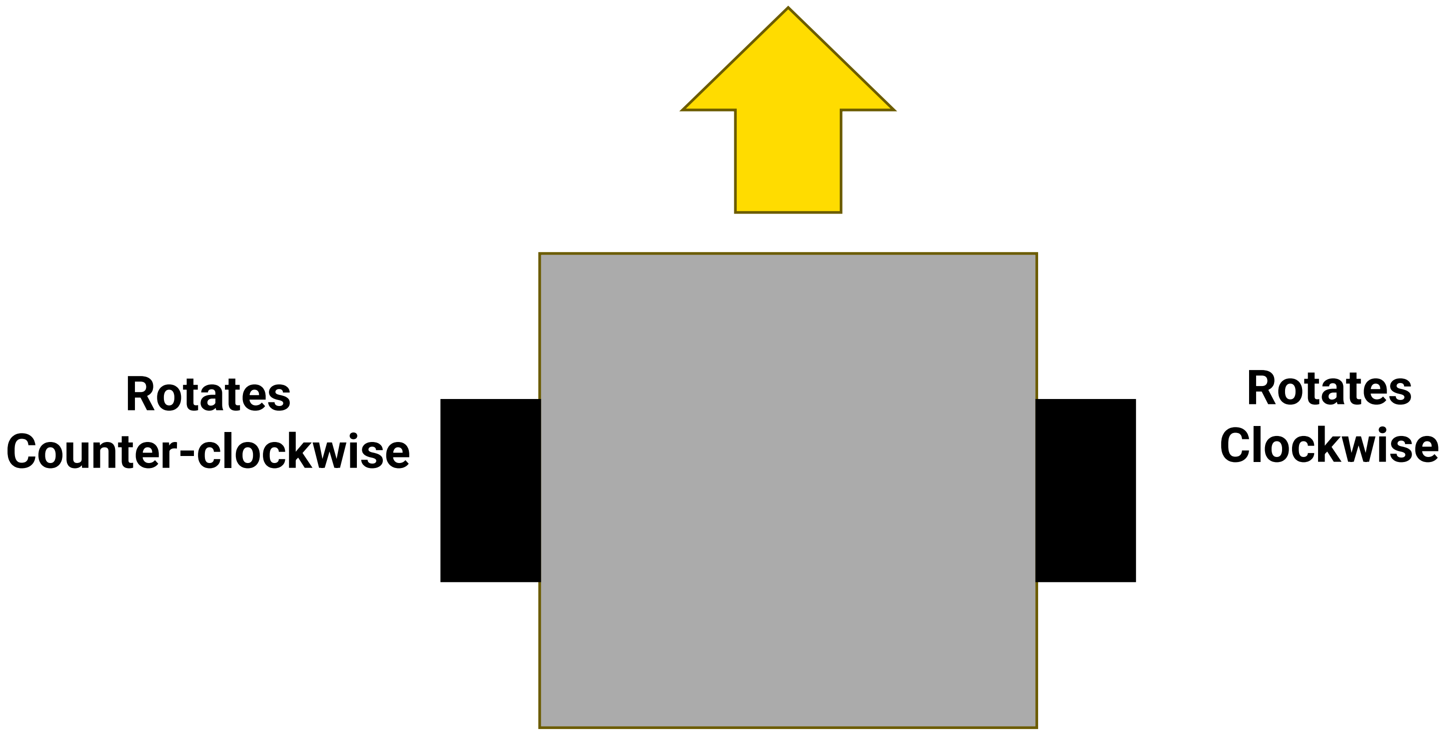
May not be the best solution for a differential robot as not all actions will result in the same output over time.



$$\boxed{\begin{array}{l} s_1 = s_2,\ s_{1R} = s_{2R},\ s_{1L} = s_{2L} \\ x_1 \neq x_2,\ y_1 \neq y_2 \end{array}}$$

Note: In the example above both robots conducted the same total quantity of wheel rotations however because of a different order they have different final poses.

Differential Drive Wheel Directions



The wheels and by extension the encoders update inverse to each other. Make sure to check and account for this.

IMU odometry

UNSW SYDNEY

Acceleration to Pose

Acceleration $\xrightarrow{\text{Integrate with respect to time}}$ **+= Velocity** $\xrightarrow{\text{Integrate with respect to time}}$ **+= Position**

IMU odometry may not always be a great solution for a long period of time. Significant drift can accumulate.

Lab tasks

Lab

Labs will happen in your project group this week. However try to work as a team of two.